

SENSOR FUSION CHALLENGE PROPOSAL

GRIP

Ground Recognition Intelligence Platform

Multi-Modal Proprioceptive Terrain Sensing for Wheeled Robots

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1 Executive Summary

Mobile ground robots cannot distinguish between different surfaces using conventional sensors. A delivery robot does not know when it transitions from tile to carpet, and an outdoor rover cannot tell concrete from gravel. This surface blindness causes navigation errors, inefficient energy use, and operational failures.

GRIP combines three sensor types—inertial measurement, acoustic sensing, and chassis vibration—into a unified surface classifier. The system runs on Arduino Uno Q with Edge Impulse, providing real-time classification across seven surface types: carpet, wood, tile, concrete, asphalt, gravel, and grass.

The core technical challenge is discrimination. Smooth wood and smooth tile produce similar vibration signatures while carpet and grass both create muffled acoustic patterns. Single sensors fundamentally cannot resolve these ambiguities. Multi-modal fusion enables discrimination by combining complementary information streams where each sensor resolves ambiguities left by the others.

A comprehensive ablation study will train seven models across single, dual, and full fusion configurations to demonstrate that fusion is essential for reliable surface classification. Target performance is 80% overall accuracy with per-class accuracy exceeding 75% and inference latency below 150 milliseconds.

GRIP provides universal terrain sensing capability applicable to any wheeled robot platform, enabling navigation adaptation, energy optimization, and terrain-aware control without requiring vision systems or external reference infrastructure.

2 The Problem

2.1 Surface Blindness in Mobile Robotics

Ground mobile robots navigate using wheel encoders and inertial measurement units. These sensors measure motion and orientation but provide no information about surface properties. A robot driving from tile to carpet experiences dramatically different traction, resistance, and mechanical loading yet has no awareness of this change.

This surface blindness creates three critical operational failures. Navigation errors accumulate when wheel encoders count rotations rather than actual distance traveled. Wheels slip on smooth surfaces during acceleration, causing odometry to overestimate distance. Wheels sink into compliant surfaces like thick carpet, causing odometry to underestimate distance. Position estimates drift from reality without external reference systems like GPS or visual landmarks, which may be unavailable or unreliable indoors.

Energy waste results from fixed control parameters applied regardless of surface type. Excessive power on smooth floors wastes energy while insufficient power on rough terrain causes motor stall or failed traversal. Energy consumption varies dramatically across surfaces, but robots cannot measure or adapt to this variation, reducing operational range and causing premature battery depletion.

Behavioral adaptation failures occur because surfaces impose different constraints on safe operation. Smooth surfaces enable high speeds but risk uncontrolled sliding due to low friction. Rough surfaces require reduced speed to prevent mechanical damage from high resistance. Surface transitions create sudden traction shifts that cause instability. Robots must either operate conservatively everywhere, sacrificing efficiency, or aggressively everywhere, sacrificing reliability.

2.2 Why Existing Solutions Fail

Vision-based classification proves computationally expensive, requiring GPU or high-end processors unsuitable for embedded platforms. Visual systems fail with texture-similar surfaces where wood looks like tile and new concrete resembles old asphalt. Lighting dependence causes failures in darkness, shadows, or glare. Vision sees ahead prospectively but not beneath wheels where contact actually occurs.

Wheel slip detection operates reactively rather than proactively, detecting problems only after they occur. The approach suffers from circular reasoning by using motion sensors to infer motion errors. Slip detection cannot distinguish between similar-friction surfaces that behave differently under varying conditions.

Single proprioceptive sensors each fail to provide complete classification. Inertial measurement units alone cannot distinguish smooth surfaces since wood, tile, and concrete all produce low vibration. Microphones alone cannot distinguish acoustically similar surfaces where carpet and grass both produce muffled signatures. Accelerometers alone provide texture information but no material identification capability.

2.3 The Fundamental Gap

No existing system provides self-contained, real-time surface classification using proprioceptive sensing. Robots need to know what they are driving on, not just where they are or how fast they are moving. GRIP addresses this gap through multi-modal sensor fusion, combining vibration, acoustic, and texture sensing to create surface signatures that are unique and reliably detectable.

3 The Solution

3.1 System Architecture

GRIP fuses three heterogeneous Qwiic-compatible sensors into a unified surface classifier running entirely on Arduino Uno Q with Edge Impulse. The system provides real-time surface classification across seven types plus unknown with sub-150ms latency and complete on-device processing without external dependencies.

3.2 The Sensor Suite

Three complementary sensors provide orthogonal information dimensions for surface characterization. The ICM-20948 9-Axis IMU, built into Arduino Uno Q, measures robot body vibration induced by wheel-surface contact at 100Hz sampling. Different surfaces create different vibration patterns through mechanical coupling, with the sensor capturing vibration amplitude as an energy measure of surface roughness, frequency spectrum in the 5-200Hz band for texture signature, and vertical versus horizontal vibration ratio indicating surface compliance. The IMU distinguishes rough from smooth surfaces and compliant from rigid surfaces but fails to distinguish smooth wood from smooth tile since both produce identical low vibration.

The INMP441 MEMS Microphone as a Qwiic module captures acoustic signatures of rolling wheel-surface contact at 16kHz sampling. Wheel deformation and surface interaction create material-specific acoustic emissions, with the microphone mounted on the underside of the robot chassis 2-3 centimeters from ground and facing downward to minimize ambient noise. The sensor provides acoustic spectrum revealing material resonances and damping characteristics, spectral brightness distinguishing hard bright surfaces from soft muted surfaces, and transient events like gravel pops,

tile clicks, or grass rustling. The microphone distinguishes hard from soft materials and acoustically reflective from absorptive surfaces but fails to distinguish acoustically similar surfaces like carpet versus grass, both heavily damped.

The ADXL345 Chassis Accelerometer as a Qwiic module measures high-frequency vibration near the wheel-axle interface at 400Hz sampling, higher than the IMU to capture texture-induced vibration. Direct mechanical coupling to the wheel captures micro-texture before attenuation through the chassis, with rigid mounting near the wheel axle or motor housing. The sensor provides high-frequency texture signature in the 50-500Hz band, impact sharpness distinguishing discrete impacts from continuous vibration, and micro-texture details invisible to the body-mounted IMU. The chassis accelerometer distinguishes fine texture differences and impact-based versus continuous vibration but fails to distinguish material composition, providing texture without material identification.

3.3 Why These Three Sensors

Each sensor provides information along a different axis. The IMU captures the mechanical energy dimension through vibration amplitude and frequency. The microphone captures the acoustic material dimension through resonance and damping. The chassis accelerometer captures the texture dimension through micro-scale surface structure. This three-dimensional feature space enables discrimination between surfaces that are indistinguishable in any single dimension. The sensors are complementary rather than redundant, with each resolving ambiguities left by the others.

3.4 Fusion Architecture

All three sensors sample continuously during robot motion, collecting synchronized 2-second windows of data: 200 IMU samples at 100Hz, 32,000 microphone samples at 16kHz, and 800 chassis accelerometer samples at 400Hz.

Edge Impulse performs specialized feature extraction on each sensor stream. The IMU generates approximately 20 features including spectral peaks in the 5-200Hz band, RMS amplitude for overall vibration energy, spectral centroid for dominant frequency, and zero-crossing rate for texture roughness. The microphone produces approximately 25 features including MFCCs for material signature, spectral rolloff for brightness measure, spectral flatness for tonal versus noisy content, and peak frequencies in the contact noise band. The chassis accelerometer contributes approximately 12 features including high-frequency spectral energy in the 50-500Hz band, impact rate and sharpness, kurtosis measuring impulsiveness, and frequency band ratios.

Individual feature vectors concatenate into a unified 57-dimensional feature vector representing the complete multi-modal signature of the surface. A compact neural network processes the fused features with input layer accepting 57 features, three hidden layers of 48, 32, and 16 neurons using ReLU activation, and output layer with 8 neurons using softmax activation for one class per surface type. The network trains using supervised learning on labeled surface data with cross-validation to prevent overfitting.

Classification results are provided through multiple interfaces. Serial output at 115200 baud provides CSV format with surface type, confidence, and inference time. I2C slave mode enables register-based access for embedded integration. GPIO pins provide binary encoding of surface type with 3 pins supporting 8 classes.

3.5 Why Fusion is Essential

Surface discrimination requires fusion because single sensors fundamentally cannot resolve all ambiguities. Wood versus tile presents identical vibration signatures that cause IMU failure, but

wood produces acoustically muted signatures while tile produces bright signatures that enable audio discrimination. Carpet versus grass presents acoustically muffled signatures that cause audio failure, but carpet shows low vibration while grass shows medium-high vibration that enables IMU discrimination.

No single sensor can distinguish all pairs of surfaces. Multi-modal fusion resolves all ambiguities by leveraging complementary information. The ablation study will demonstrate this necessity by training seven models: three single-sensor configurations using IMU only with 20 features, audio only with 25 features, or chassis accelerometer only with 12 features; three dual-sensor combinations using IMU plus audio with 45 features, IMU plus chassis with 32 features, or audio plus chassis with 37 features; and one full fusion model using all three sensors with 57 features. This progression will quantify how fusion enables reliable classification where individual sensors cannot.

4 Success Criteria

4.1 Classification Performance

The model will classify surfaces into 8 discrete categories in real-time: carpet for thick pile residential or commercial flooring, wood for smooth hardwood or laminate flooring, tile for ceramic or stone with grout lines, concrete for smooth finished sidewalk or floor, asphalt for paved road or parking lot, gravel for small to medium loose stones, grass for natural lawn or turf, and unknown for low confidence or novel surfaces. For each classification, the model outputs surface type as enumerated class, confidence score from 0-100%, and inference time in milliseconds.

Target performance requires overall accuracy exceeding 80% on held-out test data for the 7-class problem excluding unknown, per-class accuracy exceeding 75% for each individual surface type, and false classification rate below 20% which is acceptable for sensor applications providing advisory data.

System performance targets inference latency below 150 milliseconds per classification on Arduino Uno Q, model size under 400KB to fit microcontroller flash memory, and update rate classifying every 1-2 seconds during robot operation.

The ablation study will quantify fusion advantage across all seven models, demonstrating the progression from single-sensor limitations through dual-sensor partial resolution to full fusion reliability. Validation will include confusion matrices showing classification performance across all surface pairs, ablation study results demonstrating fusion improvement over single sensors, live demonstration video showing real-time classification during robot operation, and cross-validation on held-out test data not used during training.

4.2 Robustness Requirements

Performance must maintain accuracy across robot speeds from 0.2 to 2.0 meters per second, demonstrate performance degradation less than 10% with moderate ambient noise, and provide stability without rapid oscillation between classes through temporal smoothing preventing jitter.

4.3 System Integration

All three sensors must function and synchronize properly with output interfaces operational through serial, I2C, and GPIO. The mounting system must allow installation on any wheeled robot platform with continuous operation for 10 minutes or more without failure or crash.

4.4 What Success Enables

A successful GRIP system provides robots with continuous surface awareness enabling navigation systems to adjust speed and path planning based on terrain, control systems to modulate motor parameters for efficiency, energy management systems to estimate remaining range, SLAM systems to create semantic maps with surface annotations, and any robotic application requiring terrain-adaptive behavior. The model's output is advisory data for decision-making systems rather than direct control commands.

5 Timeline

The first two weeks focus on hardware integration and indoor data collection. Sensors mount to the robot platform while initial data collection captures indoor surfaces including carpet, wood, and tile. Initial Edge Impulse model training begins with this indoor dataset to validate the sensing approach and feature extraction pipeline.

Weeks three and four address outdoor data collection and comprehensive model training. Data collection expands to outdoor surfaces including concrete, asphalt, gravel, and grass across varied environmental conditions. The full fusion model trains on the complete dataset while optimization for Arduino Uno Q ensures real-time inference performance and memory constraints are met.

Testing and validation consume weeks five and six. The ablation study trains all seven models for rigorous performance comparison across single-sensor, dual-sensor, and full fusion configurations. Comprehensive testing evaluates performance across speed ranges from slow traversal to maximum velocity and across varied conditions including different surface conditions and ambient noise levels. Model refinement addresses identified weaknesses through iterative training and feature engineering.

Final integration and documentation occupy weeks seven and eight. Complete system integration verifies end-to-end functionality from sensor acquisition through classification output across all interface types. Demonstration video captures system operation under realistic conditions showing real-time classification during varied surface traversal. Comprehensive documentation packages for competition submission with full technical specifications, ablation study results, confusion matrices, and validation data.

6 Broader Impact

6.1 Applications

Delivery robots benefit from mixed indoor-outdoor navigation capability, avoiding getting stuck on surface transitions between smooth floors and rough terrain. Autonomous navigation becomes more reliable when robots understand surface constraints and can plan paths accordingly. Home robots can adjust cleaning modes automatically, modulating vacuum suction based on carpet versus hard floor detection and avoiding mopping operations on carpeted areas.

Warehouse AGVs optimize routing based on floor types, selecting paths that minimize energy consumption or traversal time. Energy management improves substantially when motor control parameters adapt to surface mechanical properties. Inspection robots maintain accurate localization across varied surfaces by detecting surface transitions and adjusting odometry correction factors accordingly.

Agricultural robots detect field boundaries automatically through surface classification, distinguishing paved roads from grass boundaries from cultivated soil. Traversal behavior adapts to terrain type, with speed and control parameters optimized for the detected surface.

6.2 Technical Contributions

The project demonstrates multi-modal fusion for proprioceptive terrain sensing, proving that edge AI can enable complex environmental perception without vision systems. The design establishes plug-and-play sensor module architecture for robotics where standard interfaces enable drop-in installation on diverse platforms. GRIP provides a practical alternative to vision-based terrain classification, operating in complete darkness with significantly lower computational requirements suitable for embedded platforms. The rigorous ablation study validates fusion necessity quantitatively, establishing frameworks for future multi-modal sensing projects. An operational framework emerges for proprioceptive environmental sensing emphasizing complementary sensor selection, real-time edge processing, and advisory output for higher-level decision systems.

7 Conclusion

GRIP demonstrates that multi-modal sensor fusion enables surface classification capabilities impossible with single sensors. By combining IMU, acoustic, and chassis vibration sensing through edge AI, the system provides real-time terrain intelligence for wheeled robots operating across indoor and outdoor environments.

The project addresses Sensor Fusion Challenge criteria through three heterogeneous sensors providing complementary information along mechanical, acoustic, and texture dimensions; meaningful fusion resolving ambiguities present in single-sensor data where smooth surfaces are vibrationally identical and soft surfaces are acoustically similar; complete edge processing on Arduino Uno Q without external dependencies or cloud computation; rigorous ablation validation quantifying fusion advantage across seven model configurations; and practical application as universal sensing module for robotic platforms from delivery robots to agricultural systems.

GRIP transforms multiple weak signals into reliable terrain understanding, enabling robots to adapt behavior to their physical environment. Surface awareness is fundamental to robust autonomous operation, yet conventional sensors provide only motion and orientation without environmental context. Multi-modal proprioceptive fusion bridges this gap, demonstrating that robots can understand what they drive on through intelligent sensor combination and edge AI processing.